An unconstrained Euclidean parametrisation of phylogenetic tree space

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Abstract

Phylogenetic trees combine a discrete topology with continuous branch lengths, which makes them difficult to work with. We introduce a recursive contour-grafting construction that encodes both components using unconstrained real-valued coordinates. After fixing an order of the leaves, each new leaf is attached at a point selected by a contour traversal of the tree constructed so far. Moving a graft coordinate continuously along the contour enables changes of topology when internal branches contract to zero. For trees with n labelled leaves, we apply the construction to rooted equidistant trees, rooted trees with unrestricted branch lengths, and unrooted trees, obtaining parametrisations of dimensions $n-1$, $2n-2$, and $2n-3$, respectively. We show that the resulting construction functions are surjective and continuous onto the corresponding spaces of phylogenetic trees. We also determine the number of representations of a generic tree and calculate the Jacobian of each parametrisation. These results provide explicit change-of-variables formulae between Euclidean coordinates and the phylogenetic tree spaces, allowing probability distributions on tree space to be represented in unconstrained Euclidean coordinates, and thus paving the way for the application of methods previously unexploited in phylogenetics.

1 Introduction

Phylogenetic trees are typically represented as the combination of a discrete topology and continuous branch lengths (Paradis 2006). This representation is natural, but it makes tree space difficult to explore using methods designed for vectors in a fixed Euclidean space. In particular, changing topology ordinarily requires purpose-built operations such as nearest-neighbour interchange (NNI) or subtree pruning and regrafting (SPR), whose performance can depend strongly on the design and tuning of the proposals (Höhna and Drummond 2012; Zhang et al. 2020).

A geometric description of tree space is provided by the Billera–Holmes–Vogtmann (BHV) space (Billera et al. 2001). Each tree topology is represented by an orthant of branch-length coordinates, and different orthants meet where one or more internal branches have length zero. This construction makes clear that topology changes can be viewed continuously: an edge belonging to one topology contracts before an edge belonging to another topology grows. The resulting space is nevertheless stratified, and computational methods that move between its orthants generally require specialised treatment (Dinh et al. 2017).

Here we develop a different representation in which both topology and branch lengths are encoded by a single vector of unconstrained real parameters. The construction is recursive. After fixing an arbitrary order of the leaves, each new leaf is attached to the tree formed by the preceding leaves. Its graft point is selected by following a contour traversal of the current tree, so that continuous movement of a real-valued coordinate moves the graft point along branches and through the tree nodes. In this way, changes of topology arise without introducing a separate discrete topology variable.

We apply this contour-grafting construction to rooted equidistant trees, rooted trees with unrestricted branch lengths, and unrooted trees. Our aim is to describe the resulting parametrisations and their geometric and measure-theoretic properties, including their behaviour at topology boundaries and the relationship between Euclidean coordinates and standard local coordinates on tree space. We focus on the theory of the parametrisations rather than on any application they enable.

The remainder of the article is organised as follows. Section 2 introduces the recursive contour-grafting construction. Sections 3–5 apply it to rooted equidistant, rooted unrestricted, and unrooted trees, respectively. Section 6 compares the three parametrisations. Finally, Section 7 provides numerical validation.

2 Recursive contour-grafting construction

All our parametrisations are constructed recursively using the same basic principle. We consider an arbitrary but fixed order for the leaves. The tree is constructed by adding each leaf one by one to the tree built so far. Let T_i denote the rooted tree formed by the first i leaves and let $L(T_i)$ be the sum of the branch lengths in the tree T_i. The tree T_1 consists only of the first leaf. When adding each leaf i beyond the first one, a graft point x_i is selected on or above T_{i-1} corresponding to the point where the new leaf is attached. This process is repeated for all leaves from the second one ($i = 2$) to the last one ($i = n$), each time adding to the tree T_{i-1} built previously.

The location of a graft point x_i is parametrised using a variable $a_i \in \mathbb{R}$. If $0 \leq a_i \leq 2L(T_{i-1})$ then a_i corresponds to the distance travelled along the contour traversal of T_{i-1} before reaching the graft point. Starting at the root, the traversal follows each branch once away from the root and once back towards it, visiting the first child of each internal node and then the second. Let $|T|$ denote the set of points on the tree T. The process of finding this graft point on a tree T using the contour traversal is denoted $\gamma_T : [0, 2L(T)] \to |T|$.

We further define the extended contour traversal as illustrated in Figure 1 and described below. If $a_i > 2L(T_{i-1})$, the grafting point is above the root of T_{i-1} by length $a_i - 2L(T_{i-1})$. Likewise, if $a_i < 0$ then the grafting point is above the root of T_{i-1} by length $-a_i$. This extended contour traversal is therefore a function $\Gamma_T : \mathbb{R} \to |T|^+$. The co-domain $|T|^+$ is obtained by adding to $|T|$ the points on a branch above the root at a distance $d \in [0, \infty)$, with zero identified with the root. The extended contour traversal is defined as:

$$\Gamma_T(a) = \begin{cases} \text{point above root by } -a & \text{if } a < 0 \\ \gamma_T(a) & \text{if } 0 \leq a \leq 2L(T) \\ \text{point above root by } a - 2L(T) & \text{if } a > 2L(T) \end{cases} \tag{1}$$

At the graft point $x_i = \Gamma_{T_{i-1}}(a_i)$, a new node is created with first child equal to what was previously below that point, and second child equal to the new branch of length ℓ_i leading to the leaf i. This convention also defines the ordering of the children used in subsequent contour traversals. If x_i lies above the current root, the segment joining the old root to x_i is added to the tree and x_i becomes the new root. If x_i coincides with an existing vertex, the operation is interpreted as the corresponding limiting construction with a zero-length branch. Thus the construction takes values in the completion

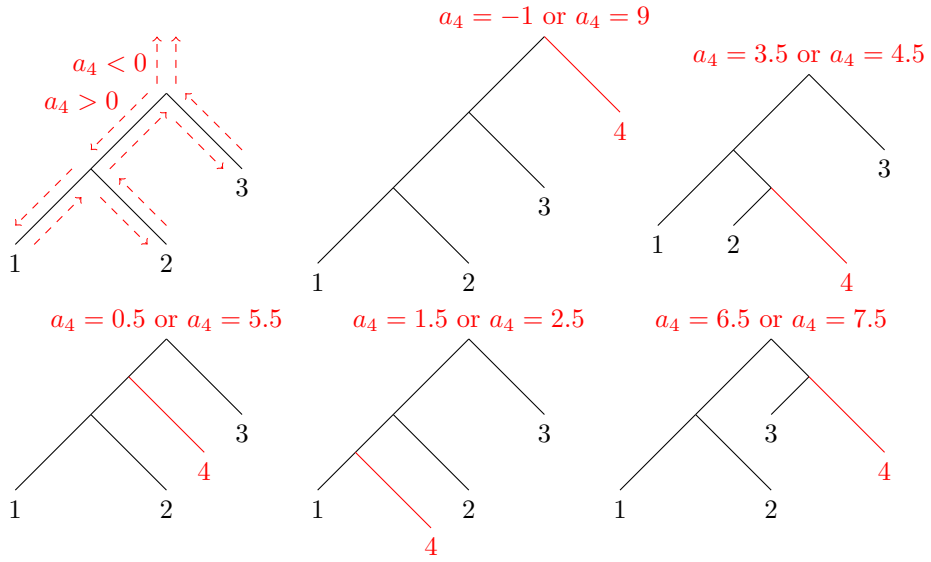


Figure 1: Illustration of the extended contour traversal. The initial tree is shown in black, with three leaves labeled 1, 2 and 3 and all four branches of length 1. Leaf 4 is added with the value of a_4 determining the graft point as shown and the new terminal branch shown in red.

of binary tree space, while almost every parameter vector produces a binary tree with strictly positive branch lengths.

The extended contour traversal is continuous because $\gamma_T(0) = \gamma_T(2(L(T))) = \text{root}(T)$. Every point in the interior of an edge of T is visited exactly twice, once in each traversal direction. A point at distance $d > 0$ above the root also has exactly two preimages, namely $-d$ and $2L(T)+d$. Tree nodes are exceptional because they may be visited more than twice; these exceptional points form the boundaries between contour regions. By considering the three cases (Equation 1) in which leaves can be added to the tree and the effect each of them has on the branch lengths we can deduce the following recursive formula for the sum of the branch lengths $L(T_i)$:

$$L(T_i) = \begin{cases} 0 & \text{if } i = 1 \\ L(T_{i-1}) + \ell_i + \max(0, a_i - 2L(T_{i-1})) + \max(0, -a_i) & \text{otherwise} \end{cases} \tag{2}$$

3 Rooted equidistant trees

We first consider phylogenetic trees with n labeled leaves that are binary, rooted and with equidistance from the root to each leaf (this property is also called ultrametric). We use the superscript E, standing for equidistant, to denote the mathematical quantities relating to this type of tree. There are $n-1$ internal nodes including the root. The topology of the tree is discrete and there are $(2n-3)!! = (2n-2)!/(2^{n-1}(n-1)!)$ unique possibilities (Felsenstein 1978). The branch lengths are continuous but not independent from each other. Since the tree is equidistant, the branch lengths are constrained so that the number of free parameters corresponds to the $n-1$ values for the heights of the internal nodes relative to the contemporaneous leaves.

The previous section introduced the recursive principle for constructing trees using the extended contour traversal for grafting, with parameters $(a_i)_{i=2,\dots,n} \in \mathbb{R}^{n-1}$ used in Equation 1. To produce an equidistant rooted tree, once a graft point x_i has been selected, the length ℓ_i of the newly created terminal branch must be equal to the height of x_i relative to the existing leaves. Specifically, for a point x on an equidistant tree T, let $h_T(x)$ denote its distance to any descendant leaf. For a point at distance d above the current root r, its distance is $h_T(x) = h_T(r) + d$. The length of the new branch is simply $\ell_i = h_{T_{i-1}}(x_i)$. This construction is detailed in Algorithm 1. The dimensionality of the parameters is $n-1$ which is the dimensionality needed to specify the $n-1$ heights of the internal nodes. A small example of applying Algorithm 1 is shown in Figure 2.

Algorithm 1 Construction of equidistant rooted trees

Require: Ordered leaves $1, \dots, n$ and parameters $C = (a_i)_{i=2,\dots,n} \in \mathbb{R}^{n-1}$
Ensure: An equidistant rooted phylogenetic tree T_n
Let T_1 consist of the single leaf 1, which is also its root
$L(T_1) \leftarrow 0$
for $i = 2, \dots, n$ **do**
    Find the grafting point $x_i = \Gamma_{T_{i-1}}(a_i)$ using Equation 1
    If x_i lies above the current root, let x_i be the new root
    Insert a new internal node at x_i, splitting its branch if necessary
    Make the subtree previously below x_i the first child of this node
    Attach leaf i as its second child by a branch of length $\ell_i = h_{T_{i-1}}(x_i)$
    Denote the resulting tree by T_i
    $L(T_i) \leftarrow L(T_{i-1}) + \ell_i + \max(0, -a_i) + \max(0, a_i - 2L(T_{i-1}))$
end for
return T_n

Let $\mathcal{C}_n^{\mathrm{E}} = \mathbb{R}^{n-1}$ denote the parameter space. Let $\mathcal{T}_n^{\mathrm{E}}$ denote the space of rooted binary equidistant trees with n labelled leaves and strictly positive branch lengths. Let $\overline{\mathcal{T}}_n^{\mathrm{E}}$ denote its completion obtained by allowing zero-length branches and identifying trees that define the same labelled metric

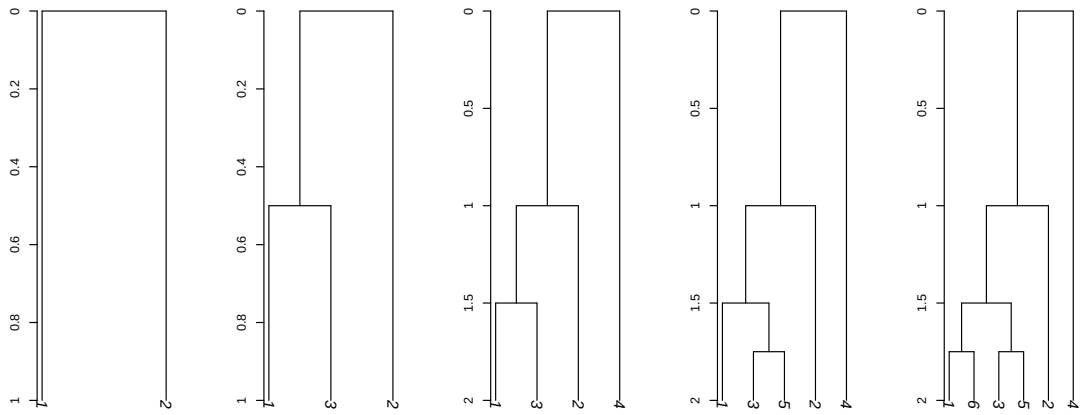


Figure 2: An example of construction of an equidistant tree using as input $n = 6$ and $(a_i)_{i=2,\ldots,n} = (-1, 1.5, -1, 2.75, 2.25)$.

tree after contraction of all zero-length edges. An element of $\mathcal{C}_n^{\mathrm{E}}$ is the vector $C = (a_i)_{i=2,\ldots,n}$, and the construction results in a unique tree in $\overline{\mathcal{T}}_n^{\mathrm{E}}$. The construction in Algorithm 1 is therefore a function $\Phi_n^{\mathrm{E}} : \mathcal{C}_n^{\mathrm{E}} \longrightarrow \overline{\mathcal{T}}_n^{\mathrm{E}}$. Almost every value of C produces a tree in $\mathcal{T}_n^{\mathrm{E}}$; the exceptional values correspond to grafting at tree nodes, including the contour endpoints.

The function Φ_n^{E} is surjective onto $\mathcal{T}_n^{\mathrm{E}}$, meaning that for any phylogenetic tree $T \in \mathcal{T}_n^{\mathrm{E}}$ there is at least one parametrisation C such that $\Phi_n^{\mathrm{E}}(C) = T$. To see this, consider any $T \in \mathcal{T}_n^{\mathrm{E}}$, remove leaf n and its pendant branch, and suppress its resulting degree-two parent. Repeating this operation produces trees with successively fewer leaves. Reversing these steps reconstructs T using the proposed grafting procedure, because every required grafting location is reached by the extended contour traversal and equidistantity uniquely determines the length of each newly added terminal branch. However, the function Φ_n^{E} is not injective as there are more than one C such that $\Phi_n^{\mathrm{E}}(C) = T$. This is a consequence of each grafting point being visited twice by the contour traversal: once as the algorithm goes down the branch and then again as it goes back up the branch. Likewise, a grafting point above the root by length x can be obtained by both $a_i = -x$ and $a_i = 2L(T_{i-1}) + x$. Each grafting point is therefore represented exactly twice if we exclude the possibility of branches of length zero. For an equidistant rooted binary tree T with strictly positive branch lengths, there are therefore $R_n^{\mathrm{E}} = 2^{n-1}$ distinct values of C such that $\Phi_n^{\mathrm{E}}(C) = T$. We call this number the redundancy of function Φ_n^{E}. This duplication of grafting coordinates is needed to achieve continuity: it allows the grafting location to move continuously through the endpoints of the contour and across changes in tree topology, without

an explicit jump between discrete topological states.

We now show that $\Phi_n^{\mathrm{E}} : \mathcal{C}_n^{\mathrm{E}} \to \overline{\mathcal{T}}_n^{\mathrm{E}}$ is continuous. Equip $\overline{\mathcal{T}}_n^{\mathrm{E}}$ with the restriction of the completed rooted Billera–Holmes–Vogtmann metric (Billera et al. 2001), including pendant-edge lengths. Within each region of parameter space in which the selected contour segments and traversal directions are fixed, all node heights and branch lengths vary continuously with the parameters. When a grafting location passes through an internal node, the separating internal branch contracts continuously to zero before the adjacent topology is entered. The two topologies therefore meet at the same boundary tree in $\overline{\mathcal{T}}_n^{\mathrm{E}}$. When the grafting location passes through a leaf, the relevant pendant segment tends continuously to zero and the two contour directions likewise meet at the same boundary tree. Similarly, the segment added above the root tends continuously to zero as a_i approaches either endpoint of the contour, namely at $a_i = 0$ or $a_i = 2L(T_{i-1})$. Applying this argument inductively at each grafting step establishes the continuity of Φ_n^{E}. Within each binary topology orthant, the heights of the internal nodes provide local coordinates. Densities on tree space are understood with respect to counting measure on topologies and Lebesgue measure in these local height coordinates.

Finally we calculate the Jacobian of $\Phi_n^{\mathrm{E}} : \mathcal{C}_n^{\mathrm{E}} \to \overline{\mathcal{T}}_n^{\mathrm{E}}$ within a binary topology. Let $\boldsymbol{h} = (h_2, ..., h_n)$ where h_i is the height of the internal node created when leaf i is grafted. Within a fixed contour region, $h_i = h_{T_{i-1}}(\Gamma_{T_{i-1}}(a_i))$. The new height depends only on $a_2, ..., a_i$, not on later parameters. Moreover,

$$\frac{\partial h_i}{\partial a_i} \in \{-1, +1\}$$

because height changes at unit rate along every fixed contour segment and along the branch above the root. Therefore the Jacobian matrix is triangular, with diagonal entries $+1$ or -1. Since the determinant of a triangular matrix is the product of the terms on its diagonal we have that the absolute Jacobian determinant is simply equal to:

$$J_n^{\mathrm{E}} = \left| \det \frac{\partial \Phi_n^{\mathrm{E}}(C)}{\partial C} \right| = \left| \det \frac{\partial (h_2, ..., h_n)}{\partial (a_2, ..., a_n)} \right| = 1 \tag{3}$$

This identity holds away from contour-region boundaries, which form a set of Lebesgue measure zero in $\mathcal{C}_n^{\mathrm{E}}$.

4 Rooted trees with unrestricted branch lengths

We now consider phylogenetic trees that are binary and rooted, with n labeled leaves. We use the superscript R, standing for rooted, to denote the mathematical quantities relating to this type of tree. There are $n-1$ internal nodes including the root. The topology of the tree is discrete and there are $(2n-3)!! = (2n-2)!/(2^{n-1}(n-1)!)$ unique possibilities (Felsenstein 1978). The branch lengths are continuous, and there are $2n-2$ branch lengths taking values in $\mathbb{R}_{>0}$ (one for each leaf and non-root internal node). Our construction is similar to the one above for equidistant trees. The main difference is that there is now another parameter $b_i \in \mathbb{R}$ for each leaf beyond the first one, that represents the length $\ell_i = \exp(b_i)$ of the branch separating the graft point (determined by a_i) and the newly added leaf i. This construction is detailed in Algorithm 2. A small example of applying Algorithm 2 is shown in Figure 3.

Algorithm 2 Construction of rooted phylogenetic trees

Require: Ordered leaves $1, \ldots, n$ and parameters $C = (a_i, b_i)_{i=2,\ldots,n} \in \mathbb{R}^{2n-2}$
Ensure: A rooted phylogenetic tree T_n
  Let T_1 consist of the single leaf 1, which is also its root
  $L(T_1) \leftarrow 0$
  for $i = 2, \ldots, n$ **do**
    Find the grafting point $x_i = \Gamma_{T_{i-1}}(a_i)$ using Equation 1
    If x_i lies above the current root, let x_i be the new root
    Insert a new internal node at x_i, splitting its branch if necessary
    Make the subtree previously below x_i the first child of this node
    Attach leaf i as its second child by a branch of length $\ell_i = \exp(b_i)$
    Denote the resulting tree by T_i
    $L(T_i) \leftarrow L(T_{i-1}) + \exp(b_i) + \max(0, -a_i) + \max(0, a_i - 2L(T_{i-1}))$
  end for
  return T_n

Let $\mathcal{C}_n^{\mathrm{R}} = \mathbb{R}^{2n-2}$ denote the parameter space. Let $\mathcal{T}_n^{\mathrm{R}}$ denote the space of rooted binary phylogenetic trees with n labelled leaves and strictly positive branch lengths. Let $\overline{\mathcal{T}}_n^{\mathrm{R}}$ denote its completion obtained by allowing zero-length branches and identifying trees that define the same labelled metric tree after contraction of all zero-length edges. An element of $\mathcal{C}_n^{\mathrm{R}}$ is the vector $C = (a_i, b_i)_{i=2,\ldots,n}$, and Algorithm 2 defines a function $\Phi_n^{\mathrm{R}} : \mathcal{C}_n^{\mathrm{R}} \longrightarrow \overline{\mathcal{T}}_n^{\mathrm{R}}$. By the same arguments as in the equidistant case, this function is continuous and is surjective onto $\mathcal{T}_n^{\mathrm{R}}$. Almost every parameter vector produces a tree in $\mathcal{T}_n^{\mathrm{R}}$, and every tree in $\mathcal{T}_n^{\mathrm{R}}$ has exactly $R_n^{\mathrm{R}} = 2^{n-1}$ preimages, which we call the redundancy of Φ_n^{R}.

As previously, we calculate the Jacobian of $\Phi_n^{\mathrm{R}} : \mathcal{C}_n^{\mathrm{R}} \to \overline{\mathcal{T}}_n^{\mathrm{R}}$ within a given binary topology. We consider this Jacobian locally, within a region of $\mathcal{C}_n^{\mathrm{R}}$ where the resulting topology, the selected contour segment for each grafting point, and its traversal direction are fixed. Let $\boldsymbol{\ell}^{(i-1)}$ denote the vector of branch

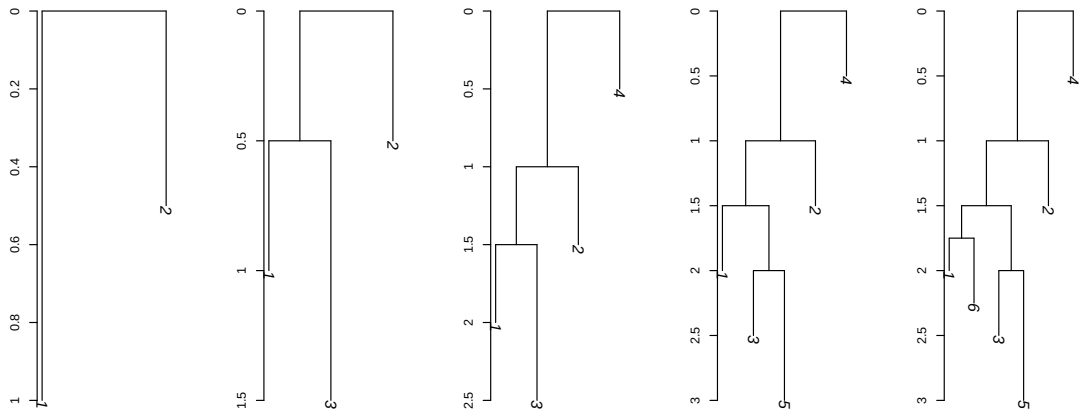


Figure 3: An example of construction of a rooted using as input $n = 6$, $(a_i)_{i=2,\dots,n} = (-1, 1.5, -1, 3, 2.25)$ and $(b_i) = i = 2, ..., n = \log(0.5, 1, 0.5, 1, 0.5)$.

lengths in T_{i-1}. If leaf i is grafted in the interior of a branch of length ℓ_j, that branch is replaced by two branches of lengths

$$u_i(\boldsymbol{\ell}^{(i-1)}, a_i) \quad \text{and} \quad \ell_j - u_i(\boldsymbol{\ell}^{(i-1)}, a_i)$$

and a terminal branch of length $\exp(b_i)$ is added. Along any fixed segment and direction of the contour traversal, u_i is affine in a_i, with

$$\frac{\partial u_i}{\partial a_i} \in \{-1, 1\}$$

After reordering the branch-length coordinates, the absolute Jacobian determinant of this grafting step is therefore

$$\left| \det \frac{\partial \boldsymbol{\ell}^{(i)}}{\partial(\boldsymbol{\ell}^{(i-1)}, a_i, b_i)} \right| = \left| \frac{\partial u_i}{\partial a_i} \right| \frac{\partial \exp(b_i)}{\partial b_i} = \exp(b_i)$$

With a suitable ordering of the branch-length coordinates, the derivative of the full recursive construction is block triangular. Consequently, the determinants contributed by successive grafting steps multiply. The same result holds when the grafting point lies above the root, since the new root branch has derivative -1 or $+1$ with respect to a_i. Applying recursively gives, for every C in which the grafting points do not coincide with a node or an endpoint of the contour:

$$J_n^{\mathrm{R}} = \left| \det \frac{\partial \Phi_n^{\mathrm{R}}(C)}{\partial C} \right| = \prod_{i=2}^{n} \exp(b_i) = \exp\left(\sum_{i=2}^{n} b_i \right) \tag{4}$$

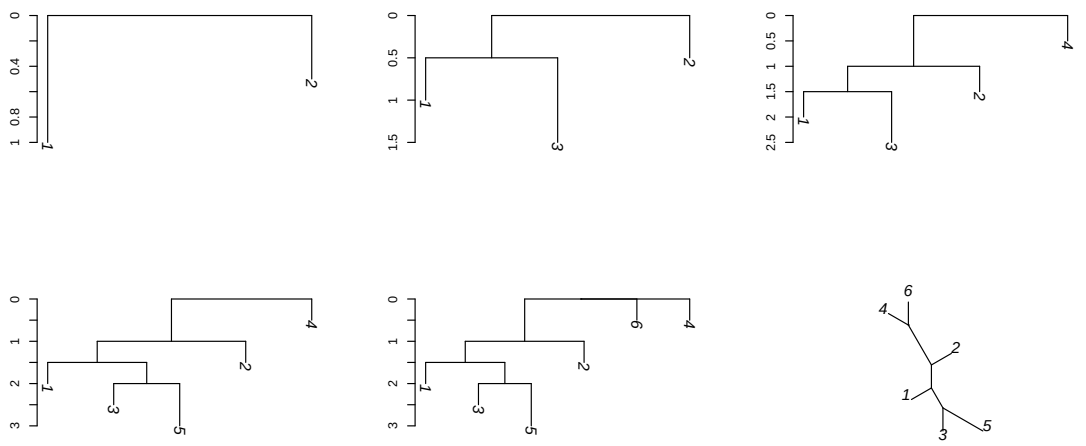


Figure 4: An example of construction of an unrooted tree using as input $n = 6$, $(a_i)_{i=2,\ldots,n-1} = (-1, 1.5, -1, 3)$, $(b_i)_{i=2,\ldots,n-1} = \log(0.5, 1, 0.5, 1)$ and $c = \log(0.5)$.

This identity holds away from contour-region boundaries, which form a set of Lebesgue measure zero in $\mathcal{C}_n^{\mathrm{R}}$.

5 Unrooted trees

We now consider unrooted binary phylogenetic trees with $n \geq 3$ labelled leaves. We use the superscript U, standing for unrooted, to denote the mathematical quantities relating to this type of tree. Such a tree has $n-2$ internal nodes, $2n-3$ branch lengths and $(2n-5)!! = (2n-5)!/(2^{n-3}(n-3)!)$ possible topologies (Felsenstein 1978). Briefly, we obtain an unconstrained parametrisation of dimension $2n-3$ by first applying the rooted construction of the previous section to the first $n-1$ leaves to produce T_{n-1}, and then grafting the last leaf n always at the temporary root of T_{n-1}, after which the root is forgotten.

Let $C = ((a_i, b_i)_{i=2,\ldots,n-1}, c) \in \mathcal{C}_n^{\mathrm{U}} = \mathbb{R}^{2n-3}$. First construct a rooted tree T_{n-1} on leaves $1, \ldots, n-1$ using Algorithm 2, with leaf 1 as the initial leaf and parameters $(a_i, b_i)_{i=2,\ldots,n-1}$. Then attach leaf n directly to the root of R by a branch of length $\exp(c)$, and then forget the root designation. The former root has degree three and therefore becomes an ordinary internal node of the resulting unrooted tree. This construction is detailed in Algorithm 3 and a small example of application is shown in Figure 4.

Algorithm 3 Parametrisation of unrooted phylogenetic trees

Require: Ordered leaves $1, \dots, n$ and parameters $C = ((a_i, b_i)_{i=2,\dots,n-1}, c) \in \mathbb{R}^{2n-3}$
Ensure: An unrooted phylogenetic tree T_n
  Run Algorithm 2 using parameters $(a_i, b_i)_{i=2,\dots,n-1}$ to obtain the rooted tree T_{n-1}
  Graft leaf n to the root of T_{n-1} by a branch of length $\exp(c)$
  Forget the root designation and denote the resulting unrooted tree by T_n
  return T_n

Let $\mathcal{T}_n^{\mathrm{U}}$ denote the space of unrooted binary phylogenetic trees with n labelled leaves and strictly positive branch lengths, and let $\overline{\mathcal{T}}_n^{\mathrm{U}}$ denote its completion obtained by allowing zero-length branches and identifying trees that define the same labelled metric tree after contraction of all zero-length edges. Algorithm 3 defines a function $\Phi_n^{\mathrm{U}} : \mathcal{C}_n^{\mathrm{U}} \longrightarrow \overline{\mathcal{T}}_n^{\mathrm{U}}$. Almost every parameter vector produces a tree in $\mathcal{T}_n^{\mathrm{U}}$; the exceptional values correspond to grafting at tree nodes, including contour endpoints.

The function Φ_n^{U} is surjective onto $\mathcal{T}_n^{\mathrm{U}}$. To see this, consider any $T \in \mathcal{T}_n^{\mathrm{U}}$. Remove leaf n and its pendant branch, and regard the resulting degree-two node as the root of the remaining tree on leaves $1, \dots, n-1$. This is a rooted binary phylogenetic tree and can therefore be represented by the function Φ_{n-1}^{R} from the previous section. Setting c to the logarithm of the removed pendant-branch length reconstructs T. Every tree in $\mathcal{T}_n^{\mathrm{U}}$ has exactly 2^{n-2} preimages. Indeed, the rooted tree on leaves $1, \dots, n-1$ has 2^{n-2} representations by the result of Section 4, while the length of the branch leading to leaf n uniquely determines c. Thus the redundancy of Φ_n^{U} is $R_n^{\mathrm{U}} = 2^{n-2}$.

The function Φ_n^{U} is continuous. Equip $\overline{\mathcal{T}}_n^{\mathrm{U}}$ with the completed unrooted Billera–Holmes–Vogtmann metric, including pendant-edge lengths. The continuity of Φ_n^{U} follows from the continuity of the function Φ_{n-1}^{R}, together with the continuity of attaching a pendant edge of length $\exp(c)$ at the root and then forgetting the root designation. At topology boundaries, zero-length edges are identified in $\overline{\mathcal{T}}_n^{\mathrm{U}}$ in the same way as in the rooted construction.

Finally, we calculate the Jacobian within a given binary topology. Use the $2n-3$ unrooted branch lengths as local coordinates. The rooted construction on leaves $1, \dots, n-1$ contributes $\exp\left(\sum_{i=2}^{n-1} b_i\right)$ to the absolute Jacobian (cf Equation 4), and the additional pendant branch has length $\exp(c)$ and derivative $\exp(c)$. With a suitable ordering of the coordinates, the full derivative is block triangular, and hence

$$J_n^{\mathrm{U}} = \left|\det \frac{\partial \Phi_n^{\mathrm{U}}(C)}{\partial C}\right| = \exp\left(c + \sum_{i=2}^{n-1} b_i\right) \tag{5}$$

almost everywhere.

The main properties of the three constructions for equidistant trees, rooted trees and unrooted trees

Tree space	Parameters	Dimensions	Redundancy	Absolute Jacobian
Rooted equidistant (E)	$(a_i)_{i=2,\dots,n}$	$n-1$	2^{n-1}	1
Rooted unrestricted (R)	$(a_i, b_i)_{i=2,\dots,n}$	$2n-2$	2^{n-1}	$\exp\left(\sum_{i=2}^{n} b_i\right)$
Unrooted (U)	$((a_i, b_i)_{i=2,\dots,n-1}, c)$	$2n-3$	2^{n-2}	$\exp\left(c + \sum_{i=2}^{n-1} b_i\right)$

Table 1: Summary of the three parametrisations.

are summarised and compared in Table 1.

6 Distribution for each tree types

We now specify a probability model for each tree type. For equidistant tree, we use the coalescent model (Kingman 1982a,b) which is the backward-in-time genealogical process for a sample of size n taken from large populations evolving under many forward-in-time models, such as the Wright-Fisher model (Wright 1931; Fisher 1930). Under this model, any pair of individuals finds a common ancestor at rate θ where θ is a scaling factor on branch lengths. The model starts with the n individuals at time zero and builds the tree backward-in-time, so that the tree is equidistant. Initially there are n individuals which means $\binom{n}{2}$ pairs and therefore the height of the first node is Exponentially distributed with rate $\binom{n}{2}\theta$. Any of the $\binom{n}{2}$ pairs is then likely to coalesce together, leaving $n-1$ ancestors. By repeating this process and noting τ_k for the duration of the time interval during which there are k ancestors, the density of a tree is obtained as:

$$p_n^{\mathrm{E}}(T) = \prod_{i=2}^{n} \exp\left(-\tau_k \binom{n}{2}\theta\right) \tag{6}$$

Under the coalescent model, the distribution of the sum of branch lengths is (Tavaré 1984):

$$p_n^{\mathrm{E}}(b) = \frac{n-1}{2\theta} \exp\left(-\frac{b}{2\theta}\right)\left(1 - \exp\left(-\frac{b}{2\theta}\right)\right)^{n-2} \tag{7}$$

The distribution of the length of a single terminal branch is (Caliebe et al. 2007):

$$p_n^{\mathrm{E}}(t) = \frac{2}{n(n-1)} \sum_{k=2}^{n} (k-1) \sum_{j=k}^{n} \left[\lambda_j \prod_{r=k, r\neq j}^{n} \frac{\lambda_r}{\lambda_r - \lambda_j} \right] \exp(-\lambda_j t) \text{ with } \lambda_j = \frac{1}{\theta} \binom{j}{2} \tag{8}$$

For rooted trees we consider a model in which all $(2n-3)!!$ topologies are equally likely and each of the $2n-2$ branch lengths ℓ_i follow an Exponential distribution with mean θ, so that:

$$p_n^{\mathrm{R}}(T) = \frac{1}{(2n-3)!!} \prod_{i=1}^{2n-2} \theta^{-1} \exp\left(-\frac{\ell_i}{\theta}\right) \tag{9}$$

Under this model a terminal branch length and the sum of branch lengths are distributed as Exponential and Gamma, respectively:

$$p_n^{\mathrm{R}}(t) = \frac{1}{\theta} \exp\left(-\frac{t}{\theta}\right) \text{ and } p_n^{\mathrm{R}}(b) = \frac{b^{2n-3}}{(2n-3)!\theta^{2n-2}} \exp\left(-\frac{b}{\theta}\right) \tag{10}$$

For unrooted trees we consider a model in which all $(2n-5)!!$ topologies are equally likely and each of the $2n-3$ branch lengths ℓ_i follow an Exponential distribution with mean θ, so that:

$$p_n^{\mathrm{U}}(T) = \frac{1}{(2n-5)!!} \prod_{i=1}^{2n-3} \theta^{-1} \exp\left(-\frac{\ell_i}{\theta}\right) \tag{11}$$

Under this model a terminal branch length and the sum of branch lengths are distributed as Exponential and Gamma, respectively:

$$p_n^{\mathrm{U}}(t) = \frac{1}{\theta} \exp\left(-\frac{t}{\theta}\right) \text{ and } p_n^{\mathrm{U}}(b) = \frac{b^{2n-4}}{(2n-4)!\theta^{2n-3}} \exp\left(-\frac{b}{\theta}\right) \tag{12}$$

7 Sampling the tree distributions

Having equiped each of the three phylogenetic tree spaces with a distribution (Equations 6, 9 and 11), we now show how to sample from these distributions using our new Euclidean parametrisations. This requires to use change-of-variables formula that has the same form in all three cases. Let $j \in \{\mathrm{E}, \mathrm{R}, \mathrm{U}\}$ denote the type of tree under study. Let Φ_n^j denote one of the three functions (Φ_n^{E}, Φ_n^{R} or Φ_n^{U}), let R_n^j be its redundancy (cf Table 1), and let $J_n^j(C)$ be its absolute Jacobian from Equations 3, 4 or 5. If p_T^j is a density on the corresponding tree space, with respect to counting measure on topologies

and Lebesgue measure in the stated local coordinates, then assigning equal mass to all equivalent representations leads (almost everywhere) to the density:

$$p_n^j(C) = \frac{p_n^j(T = \Phi_n^j(C))J_n^j(C)}{R_n^j} \tag{13}$$

Thus the parametrisations provide a similar link between a density $p_n^j(C)$ in unconstrained Euclidean space and a density $p_n^j(T)$ in phylogenetic space, for each of the three types of trees.

To sample from the distributions in Equation 13 we use an adaptive Metropolis sampler (Vihola 2011). This algorithm uses a multivariate normal distribution as the proposal kernel $q(C'|C)$ which is symmetric (meaning that $q(C'|C) = q(C|C')$), so that the acceptance probability is simply $\min(1, p_C(C')/p_C(C))$ calculated using Equation 13. The mean of the proposal kernel is the current value C and its covariance matrix is tuned to achieve an acceptance rate of 0.234 (Andrieu and Thoms 2008). For each of the three models we take $n = 5$ to generate a small example, with $\theta = 0.1$ as the the scale of the branch lengths. In each case the sampler was run for 2×10^5 iterations, with the first half discarded as burn-in and the remainder sampled every 10 iterations to produce a sample of size 10,000.

Under the coalescent model (Equation 6), all topologies are not equally likely but all ranked topologies (also known as histories) are equal likely and for $n = 5$ there are $n!(n-1)!/2^{n-1} = 180$ ranked topologies (Degnan and Salter 2005). Figure 5A shows that all 180 ranked topologies have been sampled approximately uniformly as expected. Figure 5B shows that an individual terminal branch follows the expected distribution given in Equation 8 and Figure 5C shows that the sum of branch lengths in the sampled trees follows the expected distribution given in Equation 7.

Under the rooted tree model (Equation 9) all topologies are equally likely and for $n = 5$ there are $(2n-3)!! = 105$ of them. Figure 5D shows that all of them have been sampled approximately uniformly as expected. Furthermore, Figures 5E and 5F show that terminal branch lengths and the sum of branch lengths follow the correct distributions from Equation 10. Finally, under the unrooted tree model (Equation 11) all topologies are equally likely and for $n = 5$ there are $(2n-5)!! = 15$ of them. Figure 5G shows that all of them have been sampled approximately uniformly as expected. Furthermore, Figures 5H and 5I show that terminal branch lengths and the sum of branch lengths follow the correct distributions from Equation 12.

For each of the three tree models, we used a chi-square test to confirm that the sampled topologies (Figures 5A, D and G) are as expected and we used an Anderson-Darling simple hypothesis test to

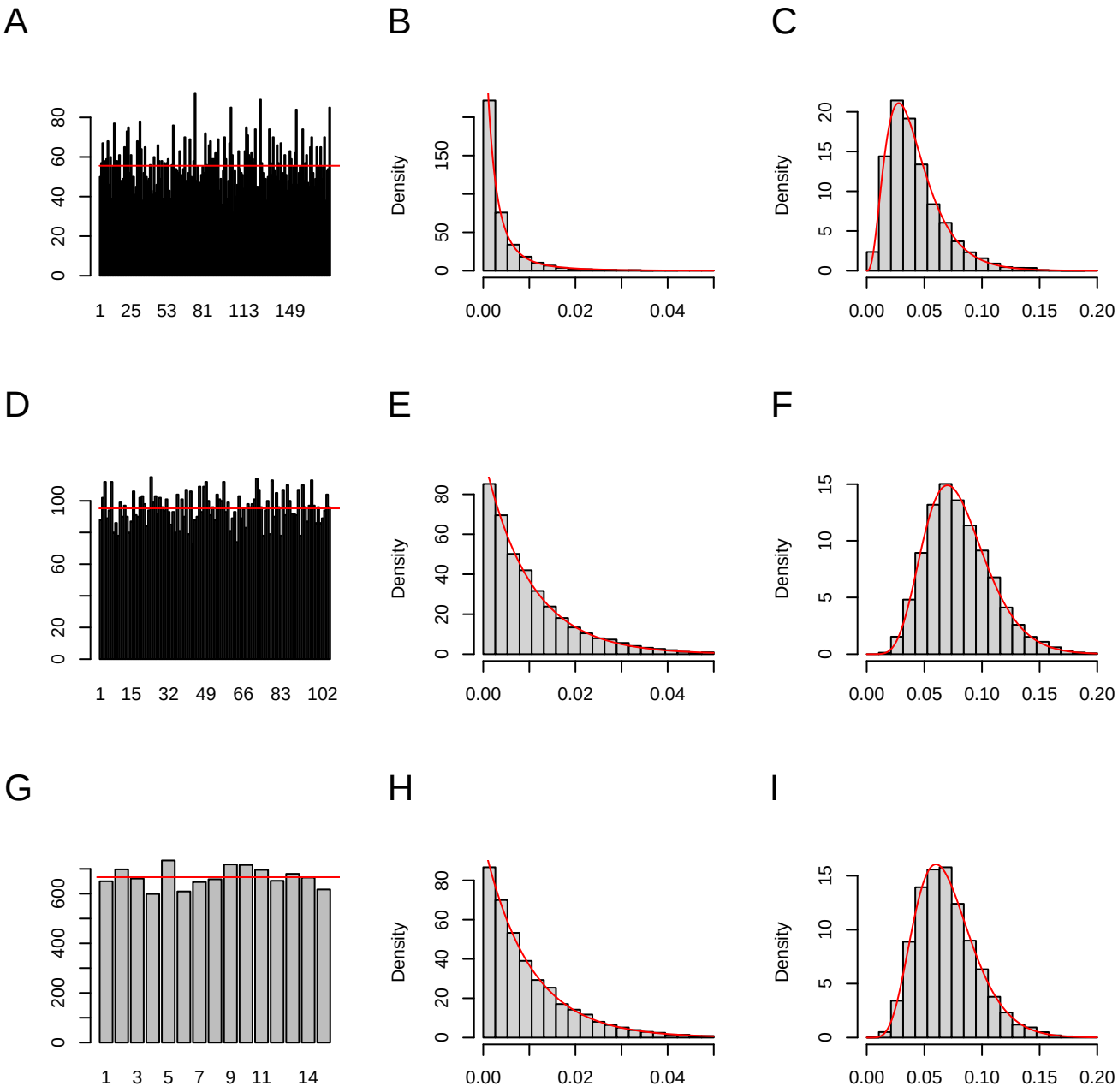


Figure 5: Sampled (histograms) and expected (lines) distributions under the equidistant tree model (A-C), rooted model (D-E) and unrooted model (F-H) for the topology (A, D, F), for a single branch length (B, E, G) and for the sum of branch lengths (C, F, H).

confirm that the sampled terminal branch lengths (Figures 5B, E and H) and sums of branch lengths (Figures 5C, F and I) are all as expected.

These results provide numerical validation of the formulas for redundancy and Jacobians of each of the three models (Table 1). They also demonstrate that it is possible to combine a real-valued sampler with the link function in Equation 13 to sample from the desired phylogenetic distributions in Equations 6, 9 and 11. This includes in particular the ability to explore the tree space over different topologies despite the Jacobian calculations applying only within each topology. Finally, this suggest the possibility of using our parametrisations within a Bayesian phylogenetic inferential framework.

8 Toy example of Bayesian phylogenetic inference

We now show a small toy example of how our new unconstrained Euclidean parametrisations can be used to perform Bayesian phylogenetic inference. After observing data D made of n aligned genetic sequences of length L, the distribution of phylogenetic trees becomes the posterior with density

$$p_n^j(C|D) \propto p_n^j(C)p(D|T = \Phi_n^j(C)) \tag{14}$$

The first term in this product is the prior density from Equation 13 and the second term is the likelihood of the data given the phylogenetic tree $T = \Phi_n^j(C)$ obtained using Algorithm 1. This likelihood can be calculated efficiently using the Felsenstein pruning algorithm (Felsenstein 1981) which is used in most phylogenetic inference tools. Many substitution models can be used to calculate this likelihood (Nascimento et al. 2017) and here we use the simplest Jukes-Cantor model (Jukes and Cantor 1969) which avoids having to infer additional parameters.

Simulation procedure. Then inference. Figure showing result, maybe including trace.

We also checked that the effective sample size was greater than 1000 for individual branch lengths (Plummer et al. 2006). The approximate effective sample size of the tree topology was also greater than 1000 (Warren et al. 2017).

9 Implementation

We implemented the methods described in this paper in a new R package entitled `ContPhylo` which is available at `https://github.com/xavierdidelot/ContPhylo` for R version 3.5 or later. This package contains all the code needed to reproduce all the results shown in this article. In particular, we implemented the Algorithms 1, 2 and 3 to construct phylogenetic trees from their Euclidean representations. We also implemented the reverse algorithms, which take as input a phylogenetic tree from one of the three types and outputs a canonical representation of this tree in the Euclidean space. There are two equivalent values of each graft coordinate a_i and the canonical representation uses the smallest positive value. We used `ape` and `phangorn` to store, manipulate and visualise phylogenetic trees (Paradis and Schliep 2019; Schliep 2011). We used the adaptive Metropolis sampler (Vihola 2011) implemented `adaptMCMC` (Scheidegger 2012). The effective sample size of continuous parameters was computed using `coda` (Plummer et al. 2006) and for the tree topology using `rwty` (Warren et al. 2017).

10 Discussion

The three constructions above provide unconstrained Euclidean parametrisations of the three types phylogenetic tree spaces considered in this article. In every case, the dimension of the parameter space equals the number of parameters needed to specify the branch lengths for a given binary topology. All three functions are surjective onto the corresponding space of binary trees with strictly positive branch lengths and extend continuously to the completed tree space obtained by allowing zero-length edges. Their parameter spaces are divided into subspaces on which the selected contour segments and traversal directions are fixed. Within each such subspace, the resulting topology is fixed and the function is continuous. The boundaries between subspaces correspond to grafting at tree nodes or contour endpoints. At these boundaries an edge contracts to zero, so adjacent binary topologies meet at the same tree in the completed space. The functions are therefore continuous globally, although they need not be differentiable across subspace boundaries. The rooted equidistant construction is the simplest case. Each added leaf requires only its graft coordinate, because equidistance uniquely determines the new terminal-branch length. The creation-ordered internal-node heights depend triangularly on the graft coordinates, with diagonal derivatives equal to $+1$ or -1, giving unit absolute Jacobian (Equation 3). Allowing unrestricted branch lengths introduces one additional log-length parameter b_i at every grafting step. Exponentiation of these parameters contributes the factors $\exp(b_i)$ to the Jacobian (Equation 4). The unrooted construction applies the unrestricted rooted construction to

leaves $1, \ldots, n-1$, attaches the distinguished leaf n to the root with length $\exp(c)$, and then forgets the root designation. It therefore removes one graft coordinate relative to the rooted case and has the minimal dimension $2n-3$, with similar Jacobian as in the rooted case (Equation 5).

In all three constructions, the redundancy of representations arise from the contour traversal. For both types of rooted tree, every graft point has two contour coordinates, so the $n-1$ successive grafts give 2^{n-1} representations. In the unrooted construction, only the rooted subtree on $n-1$ leaves is constructed by contour grafting, giving 2^{n-2} representations – the additional parameter c is uniquely determined by the length of the pendant edge leading to leaf n. These counts can change on the measure-zero set of boundary trees, where grafting points coincide with tree nodes.

The aim of Bayesian phylogenetic inference is to reconstruct a tree given some genetic data using Bayesian statistics to capture the posterior distribution of the tree (Nascimento et al. 2017). This is usually done using a Markov Chain Monte-Carlo (MCMC) to generate samples from this posterior distribution (Gilks et al. 1996). The MCMC needs to propose updates to both the branch lengths and the topology, with the latter using topology-changing moves such as nearest-neighbor interchange (NNI) or subtree pruning and regrafting (SPR). However, these moves are not easy to optimise (Höhna and Drummond 2012; Zhang et al. 2020). Conversely, many promising MCMC methods have not been applied in Bayesian phylogenetics because they can only be applied to continuous parameters. Our new unconstrained Euclidean parametrisation of phylogenetic trees provides an opportunity to use MCMC methods that had not previously been possible to use in Bayesian phylogenetics. As an example of this we used an adaptive Metropolis algorithm (Vihola 2011) to infer the phylogeny, and showed that it can sample the prior when no data is available (Figure 5) and the posterior when data is provided (Figure 6).

However, ... The focus here was on providing parametrisations that enable the use of new algorithms, not on the application of these algorithms. Implementation is in R and not optimised. Scalability not the aim but future work.

This includes in particular some adaptive methods where proposal parameters are learned during sampling (Andrieu and Thoms 2008), as well as gradient-based approaches such as the Langevin and Hamiltonian methods (Roberts and Rosenthal 1998; Neal 2011; Betancourt 2017). For example the dynamic Hamiltonian Monte Carlo (HMC) method implemented in Stan (Carpenter et al. 2017; Betancourt 2017).

Figure 6: TODO

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